

MODULE 2 - AMPLITUDE MODULATION FUNDAMENTALS

Introduction

Communication is the process of transferring information meaningfully (voice, picture, text etc.) between sender and receiver. The purpose of communication system is to transmit information bearing signals through a communication channel separating the transmitter and the receiver. The original information signal is called Baseband signal and if it is transmitted directly over a medium is referred to as Baseband Transmission. Ex: Voice telephony.

Since base band signals cannot be transmitted over long distances over the channel, the frequency of the baseband signals are modified (frequency translation) by Modulation process before transmission and at the receiver, the received signals are converted to their original form by Demodulation process.

The information signal (baseband signal) is called Modulating signal and the high frequency signal which carries information is called Carrier signal. Since the common form of carrier is a sinusoidal wave and sometimes the modulation is known as continuous wave (CW) modulation.

Modulation and Types of Modulation

Modulation is the process of changing some characteristics of a carrier wave (amplitude, phase or frequency) in accordance with the instantaneous value of modulating signal. Based on the modification of the characteristics of the carrier, there are three important types of analog modulation - (i) Amplitude Modulation (ii) Frequency Modulation and (iii) Phase Modulation.

Advantages or Need of Modulation (a) Modulation reduces the size of the antenna and Increases the range of communication (b) Modulation avoids the mixing of signals and allows multiplexing of signals (c) Modulation allows adjustments and improvement in Bandwidth (d) Modulation reduces noise and interferences and improves quality of reception

AM Concepts, Sidebands and Frequency domain:

1) Define Amplitude modulation and derive an expression for the AM signal in-terms of modulation index and plot the AM wave in Time Domain and Frequency domain.

Amplitude Modulation is type of modulation in which the information signal varies the amplitude of the carrier signal. The instantaneous value of the carrier amplitude changes in accordance with the amplitude variations of the modulating signal while the frequency and phase of the carrier signal remains constant.

Let v_m and v_c represents sinusoidal the modulating signal and carrier signal respectively and defined with proper notations as.

$$v_m = V_m \sin(2\pi f_m t) \text{ --- (1)}$$

$$v_c = V_c \sin(2\pi f_c t) \text{ --- (2)}$$

The instantaneous value of either the top or the bottom voltage envelope v_1 can be computed by using the equation $v_1 = V_c + v_m = V_c + V_m \sin(2\pi f_m t)$ ----- (3)

The instantaneous value of the modulated envelope v_2 is defined as $v_2 = v_1 \sin(2\pi f_c t)$ ----- (4)

Using equation (3) in equation (4) results

$$v_2 = [V_c + V_m \sin 2\pi f_m t](\sin 2\pi f_c t) = V_c \sin 2\pi f_c t + V_m (\sin 2\pi f_m t) (\sin 2\pi f_c t) \text{ ----- (5)}$$

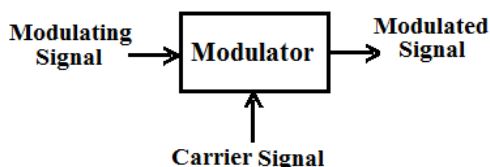
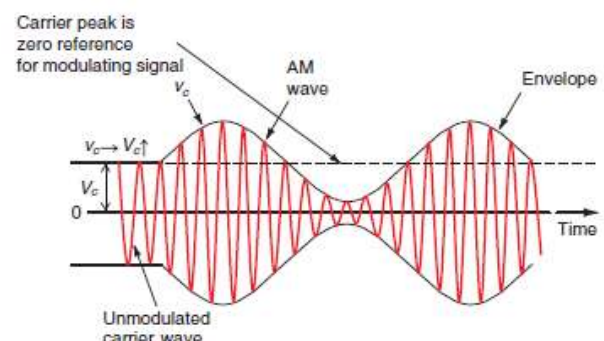


Figure shows a single frequency sine wave intelligence (modulating) signal modulating a higher-frequency carrier. The carrier frequency remains constant during the modulation process, but its amplitude varies in accordance with the modulating signal..



The envelope of the modulating signal varies above and below the peak carrier amplitude. In general, the amplitude of the modulating signal should be less than the amplitude of the carrier.

The existence of sidebands in the AM signal can be shown using the equation (5)

Using the trigonometric identity $\sin A \sin B = \frac{\cos(A-B)}{2} + \frac{\cos(A+B)}{2}$ in equation (5) results

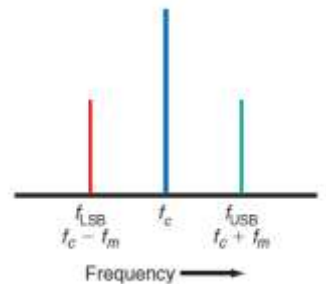
$$v_2 = V_C \sin 2\pi f_c t + \left(\frac{V_m}{2}\right) \cos 2\pi(f_c - f_m)t + \left(\frac{V_m}{2}\right) \cos 2\pi(f_c + f_m)t \quad \text{-----(6)}$$

Using the relation modulation index $m = \left(\frac{V_m}{V_C}\right)$ in the above equation, the expression becomes

$$v_2 = V_C \sin 2\pi f_c t + m \left(\frac{V_C}{2}\right) \cos 2\pi(f_c - f_m)t + m \left(\frac{V_C}{2}\right) \cos 2\pi(f_c + f_m)t \quad \text{-----(7)}$$

Eqn (7) represents the expression for AM wave in-terms of modulation index m

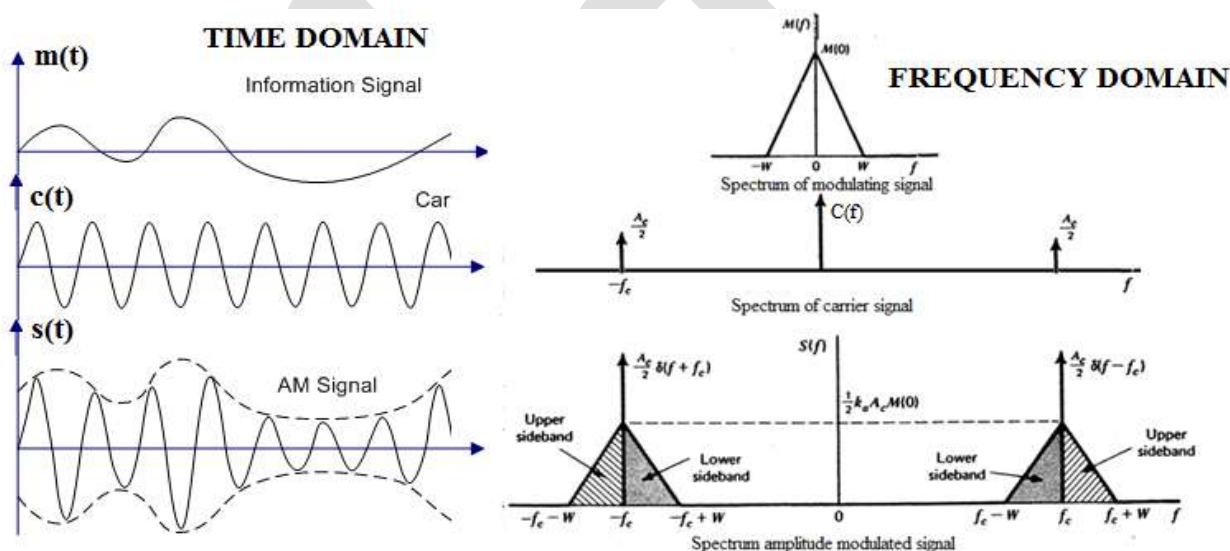
The first term represents the un-modulated Carrier signal, the second term represents the Lower Side Band (LSB) containing difference frequency ($f_c - f_m$) and last term represents the Upper Side Band (USB) containing sum frequency ($f_c + f_m$). Considering the frequency components, the Frequency Spectrum of AM signal for single tone modulation is shown in figure and the bandwidth of the AM signal is given by



$$BW_{AM} = f_{USB} - f_{LSB} = (f_c + f_m) - (f_c - f_m) = 2f_m$$

Hence the Transmission Bandwidth is equal to twice the highest frequency of the modulating signal.

NOTE- If the modulating signal occupies a bandwidth of W Hz having multiple frequency components (multi-tone signal) the frequency spectrum for multi-tone modulating signal is shown below



Modulation Index (μ) Percentage Modulation and Significance of Modulation Index

2) Define Modulation index, Percentage of modulation and explain the significance of modulation index.

Modulation Index m is defined as the ratio of the amplitude of the modulating signal to the amplitude of the Carrier wave and is a figure of merit for a modulation scheme.

$$\text{Modulation index } m = \frac{\text{Amplitude of the modulating signal}}{\text{Amplitude of the carrier signal}} = \frac{V_m}{V_c} ; (0 < m < 1)$$

Multiplying the modulation index by 100 gives the percentage of modulation.

$$\text{Percentage of Modulation } \% m = \frac{V_m}{V_c} \times 100$$

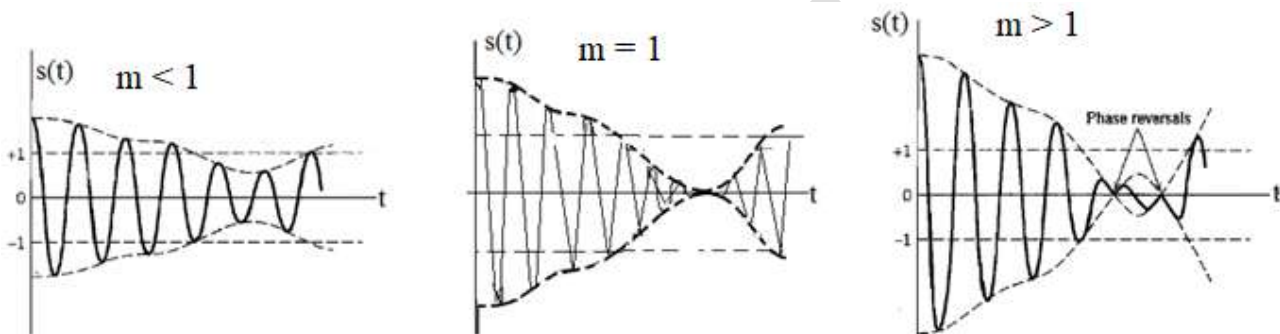
Significance of Modulation Index

Figure shows the AM waveform for various values of modulation index

If $m < 1 \rightarrow$ Under modulation. ($|V_c| > |V_m|$)

If $m = 1 \rightarrow$ Ideal Modulation or Critically damped modulation ($|V_c| = |V_m|$)

If $m > 1 \rightarrow$ Over modulation; ($|V_c| < |V_m|$) For $m > 1$; since $V_m > V_c$ phase reversal of the carrier takes place at zero crossings and this leads to cancellation of signal and results in distortion.



Over Modulation and its effects

3) What is Over Modulation ? What are its effects? How it can be minimized?

Over Modulation - When the amplitude of the information signal V_m is much greater than the amplitude of the carrier signal resulting in a condition called Over Modulation. The modulation index m is greater than 1 which causes distortion in the AM signal.

Effects of Over Modulation (a) Distortion in the modulated wave, the extent of distortion is such that results in loss of information signal due to phase reversal of the carrier signal. Distortion of voice transmissions produces garbled, harsh or unnatural sounds in the speaker. Distortion of video signals produces a scrambled and inaccurate picture on a TV screen. (b) Distortion caused by over modulation also produces adjacent channel interference.

Effects of over modulation can be minimized by using Automatic Circuits called Compression Circuits which amplify the lower-level signals and suppressing or compressing the higher-level signals. The result is a higher average power output level without over modulation.

Calculation of Modulation Index:

4) Derive an expression for Modulation Index in terms of maximum and minimum value of the AM signal.

Figure shows the Single tone AM wave and from the figure,

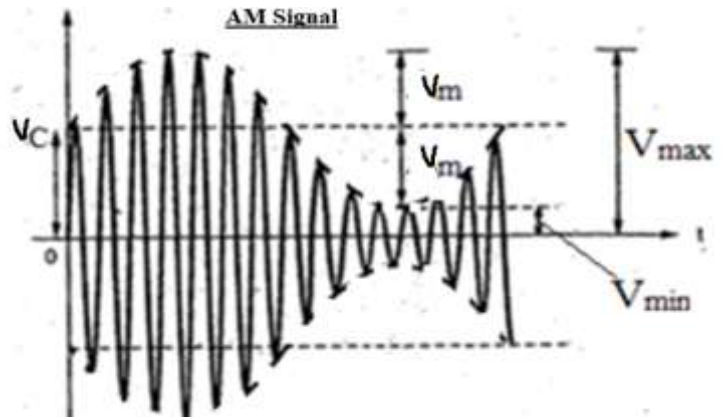
$$V_m = \frac{V_{\max} - V_{\min}}{2} \quad \text{--- (A)} \quad V_c = V_{\max} - V_m \quad \text{--- (B)}$$

Using equation (A) in equation (B)

$$V_c = V_{\max} - \frac{V_{\max} - V_{\min}}{2} = \frac{V_{\max} + V_{\min}}{2} \quad \text{--- (C)}$$

$$m = \frac{V_m}{V_c} = \frac{\frac{V_{\max} - V_{\min}}{2}}{\frac{V_{\max} + V_{\min}}{2}} = \left[\frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \right]$$

$$\frac{V_{\max}}{V_{\min}} = \frac{1 + m}{1 - m} \quad \text{--- (D)}$$



Expression for Transmitted Power of an AM wave

5) Derive an expression for total transmitted power of an AM signal.

Consider the AM signal equation

$$v_2 = V_{AM} = V_c \sin 2\pi f_c t + \left(\frac{V_m}{2} \right) \cos 2\pi (f_c - f_m) t + \left(\frac{V_m}{2} \right) \cos 2\pi (f_c + f_m) t$$

Expressing the equation in terms of RMS values

$$V_{AM} = \frac{V_c}{\sqrt{2}} \sin 2\pi f_c t + \left(\frac{V_m}{2\sqrt{2}} \right) \cos 2\pi (f_c - f_m) t + \left(\frac{V_m}{2\sqrt{2}} \right) \cos 2\pi (f_c + f_m) \quad \text{--- (a)}$$

Using the relation $m = \frac{V_m}{V_c}$ in equation (a) results

$$V_{AM} = \frac{V_c}{\sqrt{2}} \sin 2\pi f_c t + \left(\frac{m V_c}{2\sqrt{2}} \right) \cos 2\pi (f_c - f_m) t + \left(\frac{m V_c}{2\sqrt{2}} \right) \cos 2\pi (f_c + f_m) \quad \text{--- (b)}$$

The AM wave in equation (b) has three components – Unmodulated carrier, LSB and USB.

The total power of the AM wave across a standard resistor R is given by

$$P_T = P_C + P_{SB} = P_C + P_{LSB} + P_{USB} = \frac{\left(\frac{V_c}{\sqrt{2}} \right)^2}{R} + \frac{\left(\frac{m V_c}{2\sqrt{2}} \right)^2}{R} + \frac{\left(\frac{m V_c}{2\sqrt{2}} \right)^2}{R} = \frac{V_c^2}{2R} + m^2 \frac{V_c^2}{8R} + m^2 \frac{V_c^2}{8R}$$

$$P_T = \frac{V_c^2}{2R} + \frac{m^2 V_c^2}{4R} + \frac{m^2 V_c^2}{4R} = \frac{V_c^2}{2R} \left(1 + \frac{m^2}{2} \right) = P_C \left(1 + \frac{m^2}{2} \right) \quad \text{--- (c)}$$

$$\text{Where } P_C = \frac{V_c^2}{2R}; \quad P_{USB} = P_{LSB} = \left(\frac{1}{2} \right) \frac{m^2 V_c^2}{2R} = \left(\frac{m^2}{4} \right) P_C$$

$$\text{Total side band power } P_{SB} = P_{USB} + P_{LSB} = \left(\frac{m^2}{4} \right) P_C + \left(\frac{m^2}{4} \right) P_C = \left(\frac{m^2}{2} \right) P_C$$

$$P_T = P_C \left(1 + \frac{m^2}{2} \right) \quad \text{Or } m = \sqrt{2 \left(\frac{P_T}{P_C} - 1 \right)}$$

$$\text{Note: For 100\% modulation, } m = 1; \quad P_C = \frac{P_T}{\left(1 + \frac{m^2}{2} \right)} = \frac{P_T}{\left(1 + \frac{1}{2} \right)} = 0.667 P_T \quad \text{or} \quad P_T = 1.5 P_C$$

In AM wave, the 66.67% of the transmitted power is used by the Carrier signal and remaining 33.33% of the power is available with the sidebands. The graph shows the variation of power with respect to percentage modulation.

Transmission efficiency of an AM Wave

6) Derive the expression for transmission efficiency of Amplitude Modulated Wave

$$\text{Efficiency } \eta(\%) = \frac{\text{Power carried by sidebands}}{\text{Total power}} = \frac{P_{LSB} + P_{USB}}{P_T} = \frac{\left(\frac{m^2}{4} P_C\right) + \left(\frac{m^2}{4} P_C\right)}{P_C \left(1 + \frac{m^2}{2}\right)}$$

$$\eta = \left[\frac{m^2}{(2 + m^2)} \times 100 \right] \%$$

Note: For ideal modulation, i.e. $m = 1 \rightarrow \eta = \frac{1^2}{(2+1^2)} = \frac{1}{(2+1)} \times 100 \% = 33.33\%$

→ Hence the maximum transmission efficiency of AM wave is 33.33% ←

Note - Modulation Index (m) in terms of Voltages and Currents

For a constant load of $R \Omega$, the total power can be written as $P_T = P_C \left(1 + \frac{m^2}{2}\right)$

$$\text{Using } P = \frac{V^2}{R} = I^2 R \rightarrow V_T = V_C \sqrt{\left(1 + \frac{m^2}{2}\right)} \quad \text{and} \quad m = \sqrt{2 \left(\frac{V_T}{V_C}\right)^2 - 1}$$

$$\text{Also } I_T = I_C \sqrt{\left(1 + \frac{m^2}{2}\right)} \quad \text{and} \quad m = \sqrt{2 \left(\frac{I_T}{I_C}\right)^2 - 1}$$

7) Explain Modulation due to Several Signals and compute the expression for effective Modulation index due several signals.

The general expression of AM wave is $s(t) = V_C \sin 2\pi f_c t + V_m (\sin 2\pi f_m t) (\sin 2\pi f_c t)$

$$s(t) = V_C \left[1 + \left(\frac{V_m}{V_C}\right) (\sin 2\pi f_m t) \right] \sin 2\pi f_c t = V_C [1 + m (\sin 2\pi f_m t)] \sin 2\pi f_c t$$

$$s(t) = V_C [1 + m_1 (\sin 2\pi f_{m1} t) + m_2 (\sin 2\pi f_{m2} t)] \sin 2\pi f_c t$$

Using the trigonometric relation : $2 \sin A \sin B = \cos(A - B) + \cos(A + B)$ and simplifying

$$s(t) = V_C \sin(2\pi f_c t) + \frac{V_C m_1}{2} [\cos[2\pi(f_c - f_{m1})t] + \cos[2\pi(f_c + f_{m1})t]] \\ + \frac{m_2 V_C}{2} [\cos[2\pi(f_c - f_{m2})t] + \cos[2\pi(f_c + f_{m2})t]]$$

The transmission bandwidth $BW = f_{USB} - f_{LSB} = 2f_{m2} \text{ Hz}$ Where $f_{m2} > f_{m1}$

$$\text{Total power } P_T = P_C + P_{LSB1} + P_{USB1} + P_{LSB2} + P_{USB2} = \frac{\left(\frac{V_C}{\sqrt{2}}\right)^2}{R} + \frac{\left(\frac{m_1 V_C}{2\sqrt{2}}\right)^2}{R} + \frac{\left(\frac{m_1 V_C}{2\sqrt{2}}\right)^2}{R} + \frac{\left(\frac{m_2 V_C}{2\sqrt{2}}\right)^2}{R} + \frac{\left(\frac{m_2 V_C}{2\sqrt{2}}\right)^2}{R}$$

$$P_T = \frac{V_C^2}{2R} + \frac{m_1^2}{2} \left(\frac{V_C^2}{2R}\right) + \frac{m_2^2}{2} \left(\frac{V_C^2}{2R}\right) = \frac{V_C^2}{2R} \left(1 + \frac{m_1^2}{2} + \frac{m_2^2}{2}\right) = \frac{V_C^2}{2R} \left(1 + \left[\frac{m_1^2 + m_2^2}{2}\right]\right) = P_C \left[1 + \frac{m^2}{2}\right]$$

$$\text{Where } m = \sqrt{m_1^2 + m_2^2}$$

In general for n modulating signals, the overall modulation index is expressed as

$$m = \sqrt{m_1^2 + m_2^2 + \dots + m_n^2}$$

Other forms of Amplitude Modulation

8) Write Explanatory note on (a) DSB-SC and (b) SSB-SC system. Mention their benefits

(a) Double Side Band – Suppressed Carrier (DSB-SC) System

In amplitude modulation, (2/3) of the transmitted power is in the carrier, which itself conveys no information. The real information is contained within the sidebands. One way to improve the efficiency of amplitude modulation is to suppress the carrier. The result is a Double-Sideband Suppressed Carrier (DSB-SC) signal. The benefit is that no power is wasted on the carrier.

Double-sideband suppressed carrier modulation is simply a special case of AM with no carrier. The DSB-SC signal, the algebraic sum of the two sinusoidal sidebands, is the signal produced when a carrier is modulated by a single-tone sine wave information signal.

Expression for DSBSC signal

$$v_{DSB} = V_c \sin 2\pi f_c t \cdot V_m \sin 2\pi f_m t$$

$$v_{DSB} = \left(\frac{V_c V_m}{2}\right) \cos 2\pi (f_c - f_m) t + \left(\frac{V_c V_m}{2}\right) \cos 2\pi (f_c + f_m) t$$

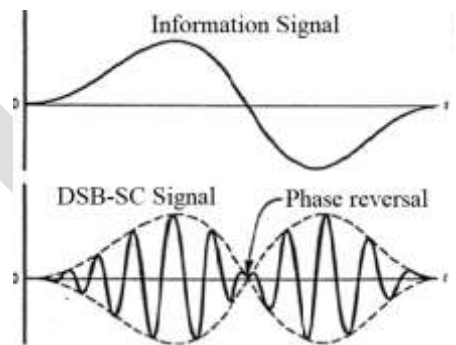
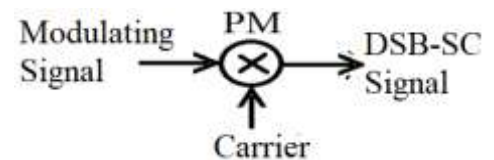
The carrier is suppressed, and the time-domain DSB signal is a sine wave at the carrier frequency, varying in amplitude as shown.. A unique characteristic of the DSB signal is the phase transitions that occur at the

lower-amplitude portions of the wave. From the frequency-domain display of a DSB signal, the spectrum space occupied by a DSB signal is the same as that for a conventional AM signal. $BW = 2f_m$

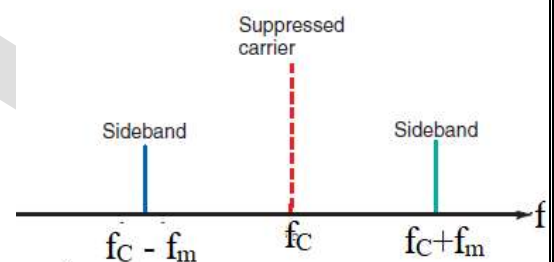
Double-sideband suppressed carrier signals are generated by a circuit called a Balanced Modulator.

Limitations: A stable local oscillator at the transmitter and receiver and also strict synchronization in frequency and phase are required

Application- One important application for DSB, is the transmission of the color information in a TV signal.



A frequency-domain display of DSB signal.

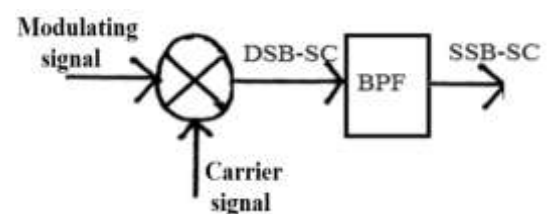


(b) Single Sideband Suppressed Carrier (SSB-SC)

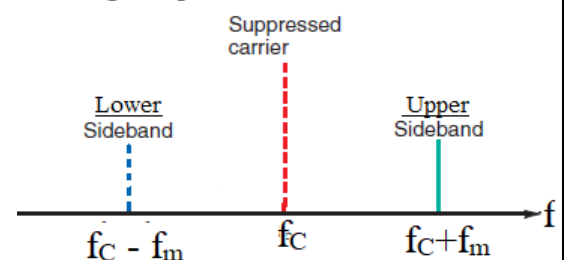
The transmission bandwidth of DSB-SC signal is shared equally by lower and upper sidebands and these sidebands are uniquely related to each other by virtue of their symmetry about the carrier frequency. Hence by suppressing the carrier and one sideband, the information can be transmitted with the other sideband which will result in the bandwidth saving. This type of signal is called Single Sideband Suppressed Carrier (SSB-SC) and the transmission of this signal is called Single Sideband Suppressed Carrier (SSB-SC) Modulation.

SSB-SC signals are obtained by passing the DSB-SC signal through an ideal BPF with cut-off frequencies f_c and $(f_c + W)$ for upper sideband [f_c and $(f_c - W)$ for Lower sideband]

The analysis of a SSB signal uses a technique known as



SSB signal spectrum with USB retained



Hilbert Transforms. Similar to a DSB signal, coherent demodulation is used to recover the SSB signal.

Advantages: (i)SSB requires half the BW as required by DSB. (ii) Due to suppression of carrier as well as one sideband, large amount of power is saved. (iii) Due to reduced BW, noise interference is minimum. (iv) SSB does not suffer from fading effects.

Limitations: A stable local oscillator at the transmitter and receiver and also strict synchronization in frequency and phase are required.

Applications: SSB transmission is used where bandwidth requirements are low. Ex: Point to point communication, Transmission of analog voice signals, telemetry, radio navigation and armature radio etc.

Signal Power and Average power in SSB

In SSB, the transmitter output is expressed in terms of peak envelope power (PEP), the maximum power produced on voice amplitude peaks.

$$PEP = \frac{[V_{rms}]^2}{R}$$

Numerical Problems

1) Suppose that on an AM signal, the $V_{\max(p-p)}$ value read from the graticule on the oscilloscope screen is 5.9 divisions and $V_{\min(p-p)}$ is 1.2 divisions. Calculate the Modulation Index. Calculate V_C , V_m , and m if the vertical scale is 2 V per division.

Solution $m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{5.9 - 1.2}{5.9 + 1.2} = \frac{4.7}{7.1} = 0.662$ or 66.2%

$$V_C = \frac{V_{\max} + V_{\min}}{2} = \frac{5.9 + 1.2}{2} = \frac{7.1}{2} = 3.55 @ \frac{2V}{\text{div}} = 3.55(2) = 7.1V$$

$$V_m = \frac{V_{\max} - V_{\min}}{2} = \frac{5.9 - 1.2}{2} = \frac{4.7}{2} = 2.35 @ \frac{2V}{\text{div}} = 2.35(2) = 4.7V$$

$$m = \left[\frac{V_m}{V_C} \right] = \left[\frac{4.7}{7.1} \right] = 0.662$$

2) An AM wave displayed on an oscilloscope has values of $V_{\max} = 4.8$ and $V_{\min} = 2.5$ as read from the graticule. What is the percentage of modulation?

Solution $m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{4.8 - 2.5}{4.8 + 2.5} = \frac{2.3}{7.3} = 0.315$ or 31.5%

3) An AM transmitter has a carrier power of 30 W. The percentage of modulation is 85 percent. Calculate (a) the total power and (b) the power in one sideband.

Solution (a) $P_T = P_C \left(1 + \frac{m^2}{2} \right) = 30 \left(1 + \frac{0.85^2}{2} \right) = 40.8 W$

(b) $P_{USB} = P_{LSB} = \left(\frac{m^2}{4} \right) P_C = \left(\frac{0.85^2}{4} \right) 30 = 5.4W$

or $P_{USB} = P_{LSB} = \left(\frac{P_T - P_C}{2} \right) = \left(\frac{40.8 - 30}{2} \right) = 5.4W$

4) An antenna has an impedance of 40Ω . An unmodulated AM signal produces a current of 4.8A . The modulation is 90 percent. Calculate (a) the carrier power (b) the total power and (c) the sideband power. (d) The transmitter experiences an antenna current change from 4.8A un-modulated to 5.1A . What is the percentage of modulation?

Solution -

(a) Using $P_c = I_c^2 R = (4.8)^2(40) = 921.6\text{ W}$

$$I_T = I_c \sqrt{\left(1 + \frac{m^2}{2}\right)} = 4.8 \sqrt{\left(1 + \frac{0.9^2}{2}\right)} = 5.7\text{A} \quad \text{(b) } P_T = I_T^2 R = (5.7)^2(40) = 1295\text{ W}$$

(c) $P_{SB} = P_T - P_c = 1295 - 921.6 = 373.4\text{ W}$; $P_{USB} = P_{LSB} = \left(\frac{P_{SB}}{2}\right) = \left(\frac{373.4}{2}\right) = 186.7\text{ W}$

(d) $m = \sqrt{2\left(\frac{I_T}{I_c}\right)^2 - 1} = \sqrt{2\left(\frac{5.7}{4.8}\right)^2 - 1} = 0.51 \text{ or } 51\%$

5) An SSB transmitter produces a peak-to-peak voltage of 178 V across a 75Ω antenna load. What is the PEP?

Solution – Given $V_{(p-p)} = 178\text{V}$; $R = 75\Omega$ $V_p = \frac{V_{(p-p)}}{2} = \frac{178}{2} = 89\text{V}$

$$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{89}{\sqrt{2}} = 62.9\text{V} \quad \text{PEP} = \frac{[V_{rms}]^2}{R} = \frac{[62.9]^2}{75} = 52.8\text{W}$$

6) An SSB transmitter has a 24-V dc power supply. On voice peaks the current achieves a maximum of 9.3 A . (a) What is the PEP? (b) What is the average power of the transmitter?

Solution- Given $V_s = 24\text{V}$; $I_m = 9.3\text{A}$ $\text{PEP} = V_s I_m = 24(9.3) = 223.2\text{ W}$

$$P_{av} = \frac{\text{PEP}}{4} \text{ to } \frac{\text{PEP}}{4} \rightarrow P_{av} = \frac{223.2}{4} = 55.8\text{ W} \text{ to } P_{av} = \frac{223.2}{3} = 74.4\text{W}$$

Average power is in the range (55.8W to 74.4 W)

Solution to Exercise Problems (Text book 2)

1) An AM wave displayed on an oscilloscope has values of $V_{max} = 4.8$ and $V_{min} = 2.5$ as read from the graticule. What is the percentage of modulation?

Solution $m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{4.8 - 2.5}{4.8 + 2.5} = \frac{2.3}{7.3} = 0.315 \text{ or } 31.5\%$

2) To achieve 75 percent modulation of a carrier of $V_c = 50\text{ V}$, what amplitude of the modulating signal V_m is needed?

Solution $V_m = m V_c = 0.75(50) = 37.5\text{V}$

3) The maximum peak-to-peak value of an AM wave is 45 V . The peak-to-peak value of the modulating signal is 20 V . What is the percentage of modulation?

Solution Given $V_{m(p-p)} = 20\text{V}$; hence $V_m = 0.5V_{m(p-p)} = 10\text{ V}$

Also given $2V_{\max(p-p)} = 45V$ or $V_{\max(p-p)} = (45/2) = 22.5V$

hence $V_{\max} = 0.5V_{\max(p-p)} = 0.5(22.5) = 11.25V$ and $V_{\min} = V_{\max} - V_m = 11.25 - 10 = 1.25V$

$$m = \left[\frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \right] = \left[\frac{11.25 - 1.25}{11.25 + 1.25} \right] = \left[\frac{10}{12.5} \right] = 0.8 \text{ or } 80\%$$

4) An AM radio transmitter operating on 3.9 MHz is modulated by frequencies up to 4 kHz. What are the maximum upper and lower side frequencies? What is the total bandwidth of the AM signal?

Solution - $f_{\text{USB}} = f_c + f_m = 3.9 + 0.004 = 3.904 \text{ MHz}$; $f_{\text{LSB}} = f_c - f_m = 3.9 - 0.004 = 3.896 \text{ MHz}$

$$BW = f_{\text{USB}} - f_{\text{LSB}} = 2f_m = 2(4) = 8 \text{ kHz}$$

5) What is the bandwidth of an AM signal whose carrier is 2.1 MHz modulated by a 1.5-kHz square wave with significant harmonics up to the fifth? Calculate all the upper and lower sidebands produced.

Solution - A square wave consists of a fundamental sine wave (of the same frequency as the square wave) and odd harmonics of the fundamental. Hence 1.5kHz square wave 3rd & 5th harmonics are 4.5kHz & 7.5 kHz.

f_m (kHz)	f_c (kHz)	$f_{\text{LSB}} = f_c - f_m$	$f_{\text{USB}} = f_c + f_m$
1.5	2100	2098.5	2101.5
4.5	2100	2095.5	2104.5
7.5	2100	2092.5	2107.5

$$f_{\text{LSB}} = 2092.5 \text{ kHz} \quad f_{\text{USB}} = 2107.5 \text{ kHz}$$

$$BW = f_{\text{USB}} - f_{\text{LSB}}$$

$$BW = 2107.5 - 2092.5 = 15 \text{ kHz}$$

6) How much power appears in one sideband of an AM signal of a 5 kW transmitter modulated by 80 %

Solution - Given $P_c = 5 \text{ kW}$ and $m = 0.8$ hence $P_{\text{LSB}} = P_{\text{USB}} = m^2(P_c/4) = 0.8^2(5/4) = 0.8 \text{ kW}$

7) What is the total power supplied by an AM transmitter with a carrier power of 2500W and modulation of 77 percent?

Solution $P_T = P_c \left(1 + \frac{m^2}{2} \right) = 2.5 \left(1 + \frac{0.77^2}{2} \right) = 3.241 \text{ kW}$

8) An AM signal has a 12-W carrier and 1.5 W in each sideband. What is the percentage of modulation?

Solution $m = \sqrt{\frac{4 P_{\text{LSB}}}{P_c}} = \sqrt{\frac{4(1.5)}{12}} = 0.707 \text{ or } 70.7 \%$

9) An AM transmitter puts a carrier of 6 A into an antenna whose resistance is 52Ω. The transmitter is modulated by 60 percent. What is the total output power?

Solution - Using $P_c = I_c^2 R = (4.8)^2(40) = 921.6 \text{ W}$

$$P_c = I_c^2 R = (6)^2(40) = 1872 \text{ W} \quad P_T = P_c \left(1 + \frac{m^2}{2} \right) = 1.872 \left(1 + \frac{0.6^2}{2} \right) = 2.209 \text{ kW}$$

10) The antenna current produced by an un-modulated carrier is 2.4A into an antenna with a resistance of 75Ω. When amplitude-modulated, the antenna current rises to 2.7A. What is the % of modulation?

Solution - Given $I_T = 2.4A$; $I_c = 2.4A$

$$m = \sqrt{2 \left[\left(\frac{I_T}{I_c} \right)^2 - 1 \right]} = \sqrt{2 \left[\left(\frac{2.7}{2.4} \right)^2 - 1 \right]} = 72.88\%$$

11) A ham transmitter has a carrier power of 750 W. How much power is added to the signal when the transmitter is 100 percent modulated?

Solution - Given $P_C = 0.75 \text{ kW}$; $m = 1$; $P_T = P_C \left(1 + \frac{m^2}{2}\right) = 0.75 \left(1 + \frac{1^2}{2}\right) = 1.125 \text{ kW}$

The power added to the signal is $P_{SB} = P_T - P_C = 1.125 - 0.75 = 0.375 \text{ kW}$

12) An SSB transmitter has a power supply voltage of 250 V. On voice peaks, the final amplifier draws a current of 3.3 A. What is the input PEP?

Solution - Given $V_{rms} = 250 \text{ V}$; $I_{rms} = 3.3 \text{ A}$ $PEP = V_{rms} I_{rms} = 250(3.3) = 825 \text{ W}$

13) The peak-to-peak output voltage of 675 V appears across a 52Ω antenna on voice peaks in an SSB transmitter. What is the output PEP?

Solution - Given $V_{(p-p)} = 675 \text{ V}$; $R = 52 \Omega$ $V_p = \frac{V_{(p-p)}}{2} = \frac{675}{2} = 337.5$
 $V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{337.5}{\sqrt{2}} = 238.68$ $PEP = \frac{[V_{rms}]^2}{R} = \frac{[238.68]^2}{52} = 1095.56 \text{ W}$

14) What is the average output power of an SSB transmitter rated at 100-W PEP?

Solution $P_{av} = \frac{PEP}{4}$ to $\frac{PEP}{3} \rightarrow P_{av} = \frac{100}{4} = 25 \text{ W}$ to $P_{av} = \frac{100}{3} = 33.33 \text{ W}$

Average power is in the range 25W to 33.33W

15) An SSB transmitter with a carrier of 2.3 MHz is modulated by an intelligence signal in the 150Hz to 4.2kHz range. Calculate the frequency range of the lower sideband.

$f_m \text{ (kHz)}$	$f_c \text{ (kHz)}$	$f_{LSB} = f_c - f_m$
0.15	2300	2299.85 kHz
4.2	2300	2295.8 kHz

Range of Lower side band is [2299.85 kHz to 2295.8 kHz]

Critical Thinking

1) A subcarrier of 70 kHz is amplitude- modulated by tones of 2.1 and 6.8kHz. The resulting AM signal is then used to amplitude-modulate a carrier of 12.5MHz. Calculate all sideband frequencies in the composite signal, and draw a frequency- domain display of the signal. Assume 100 percent modulation. What is the bandwidth occupied by the complete signal?

Stage 1			
$f_m \text{ (kHz)}$	$f_c \text{ (kHz)}$	$f_{LSB} = f_c - f_m$	$f_{USB} = f_c + f_m$
2.1	70	67.9	72.1
6.8	70	63.2	76.8

Stage 2			
$f_m \text{ (kHz)}$	$f_c \text{ (kHz)}$	$f_{LSB} = f_c - f_m$	$f_{USB} = f_c + f_m$
63.2	12500	12436.8	12563.2
68.9	12500	12431.1	12568.9
72.1	12500	12427.9	12572.1
76.8	12500	12423.2	12576.8

BW = $12576.8 - 12423.2 = 153.6 \text{ kHz}$

2) An AM signal with 100 percent modulation has an upper sideband power of 32 W. What is the carrier power.

Solution - Given $m = 1$; $P_{LSB} = P_{USB} = 32 \text{ W}$

Using the relation $P_{LSB} = P_{USB} = m^2(P_C/4) \rightarrow P_C = [4 P_{LSB}] / m^2 = [4(32)] / 1 = 128 \text{ W}$

Amplitude Modulator and Demodulator Circuits

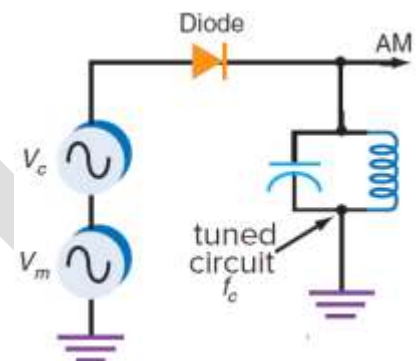
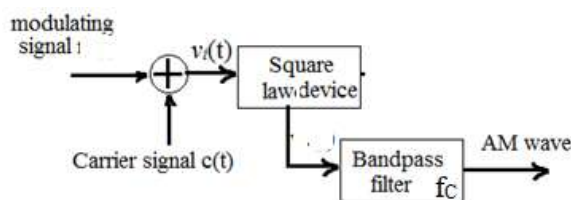
Introduction – Amplitude Modulator is an electronic circuit supplied with Modulating signal and Carrier signals as input and used to generate AM signal at the output.

Amplitude modulators are generally one of two types: low level or high level. Low-level modulators generate AM with small signals and thus must be amplified considerably if they are to be transmitted. High-level modulators produce AM at high power levels, usually in the final amplifier stage of a transmitter.

Square Law Modulator

9) With necessary schematic of square -law circuit and expressions, explain the generation of AM.

Figure shows the block diagram and circuit diagram of a square law modulator, which consists of a square law device (non linear element like diode or transistor, a summer (which adds the modulating signal and the carrier) and a bandpass filter which extracts the desired signal (can be single tuned or double tuned filter).



When a non linear device semi conductor diode is suitably biased and operated in the non linear region of its characteristic curve, the current through such device is related by the equation.

$$i = av + bv^2 \text{ -----(1) where } a \text{ and } b \text{ are constants and } a \text{ is the applied voltage } v$$

Let the modulating signal be $v_m = V_m \sin 2\pi f_m t$ and the carrier signal be $v_c = V_c \sin 2\pi f_c t$

$$\text{Hence } i = a(v_c + v_m) + b(v_c + v_m)^2 = av_c + av_m + b(v_c)^2 + b(v_m)^2 + 2abv_c v_m$$

$$i = aV_c \sin 2\pi f_c t + aV_m \sin 2\pi f_m t + b(V_c \sin 2\pi f_c t)^2 + b(V_m \sin 2\pi f_m t)^2 + 2abV_m V_c \sin 2\pi f_c t \sin 2\pi f_m t$$

$$i = [aV_c \sin 2\pi f_c t + 2abV_m V_c \sin 2\pi f_c t \sin 2\pi f_m t] + aV_m \sin 2\pi f_m t + bV_c^2 \left[\frac{1 - \cos(2\pi 2f_c t)}{2} \right] + bV_m^2 \left[\frac{1 - \cos(2\pi 2f_m t)}{2} \right] \text{ --- (A)}$$

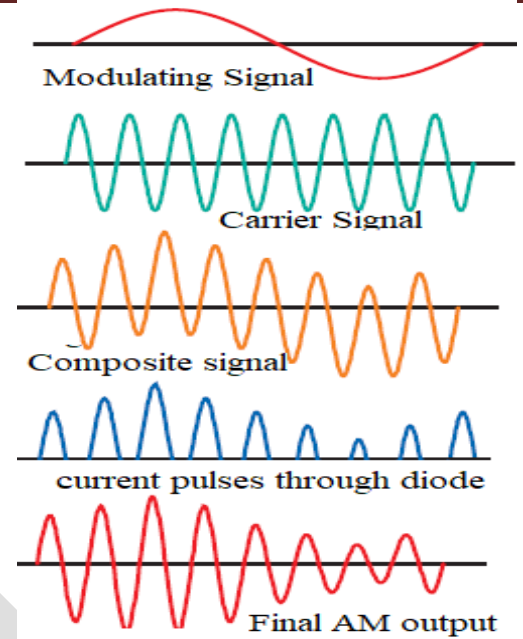
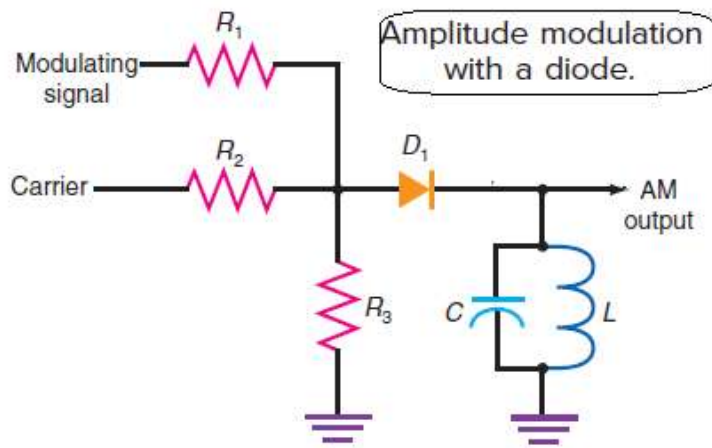
Equation (A) gives the resultant voltage developed across output of the square law device. The first term is the desired AM wave, The unwanted other is removed by a band-pass filter with centre frequency f_c and band width equal to $2f_m$.

Hence the output of the filter is $i = [aV_c \sin 2\pi f_c t + 2abV_m V_c \sin 2\pi f_c t \sin 2\pi f_m t]$ is the desired AM wave.

Low Level Amplitude Modulators

10) Explain the working of Diode Modulator circuit with all the necessary diagrams.

A Diode Modulator circuit consists of a resistive mixing circuit, a rectifier and an LC tuned resonant circuit with frequency f_c . Modulating and carrier signals are applied as inputs to the mixing circuit and the linearly mixed (algebraically added) signal appears across resistor R_3 as a result the carrier is riding on the modulating signal.

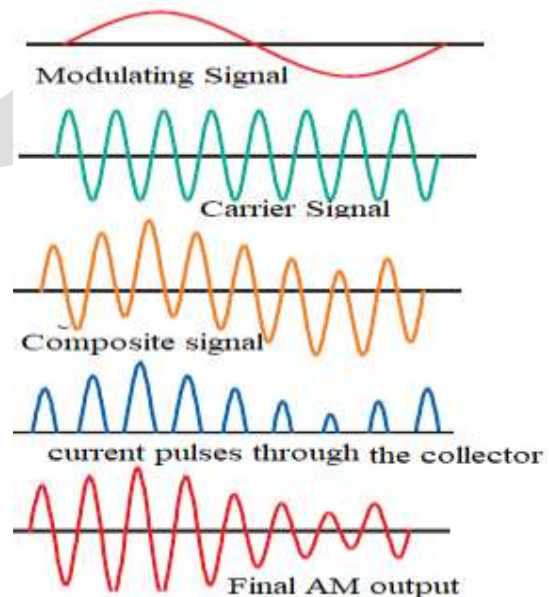
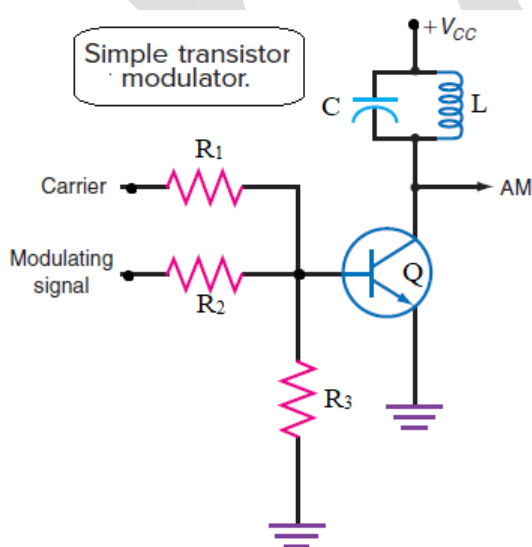


This composite signal is applied to a diode rectifier and the diode is connected in such a way that conducts only during the positive going half cycles of the applied input and during negative pulses, the diode is reverse biased and no signal passes. The current through the diode is hence a series of positive going pulses whose amplitude varies in proportional to the amplitude of the modulating signal.

These positive going pulses are applied to the parallel tuned circuit made up of L and C with resonant frequency f_c . Each time the diode conducts, a pulse of current flows through the tuned circuit (capacitor charges to the peak value of the current pulse). The coil and capacitor repeatedly exchange energy when the diode is off causing an oscillation at the resonant frequency f_c . The resulting output wave form across the tuned circuit is an AM signal.

11) Explain the working of Transistor Modulator Circuit

A Transistor Modulator circuit consists of a resistive mixing circuit, a transistor and an LC tuned resonant circuit with frequency f_c . Modulating and carrier signals are applied as inputs to the mixing circuit and the linearly mixed (algebraically added) signal appears across resistor R_3 as a result the carrier is riding on the modulating signal.



This composite signal is applied across the E-B junction of the transistor and the EB junction acts like a rectifier diode and conducts only during the positive going half cycles of the applied input and during negative pulses, the transistor off. The current through the transistor is a series of positive going pulses whose amplitude varies in proportional to the amplitude of the modulating signal and gets amplified by the transistor.

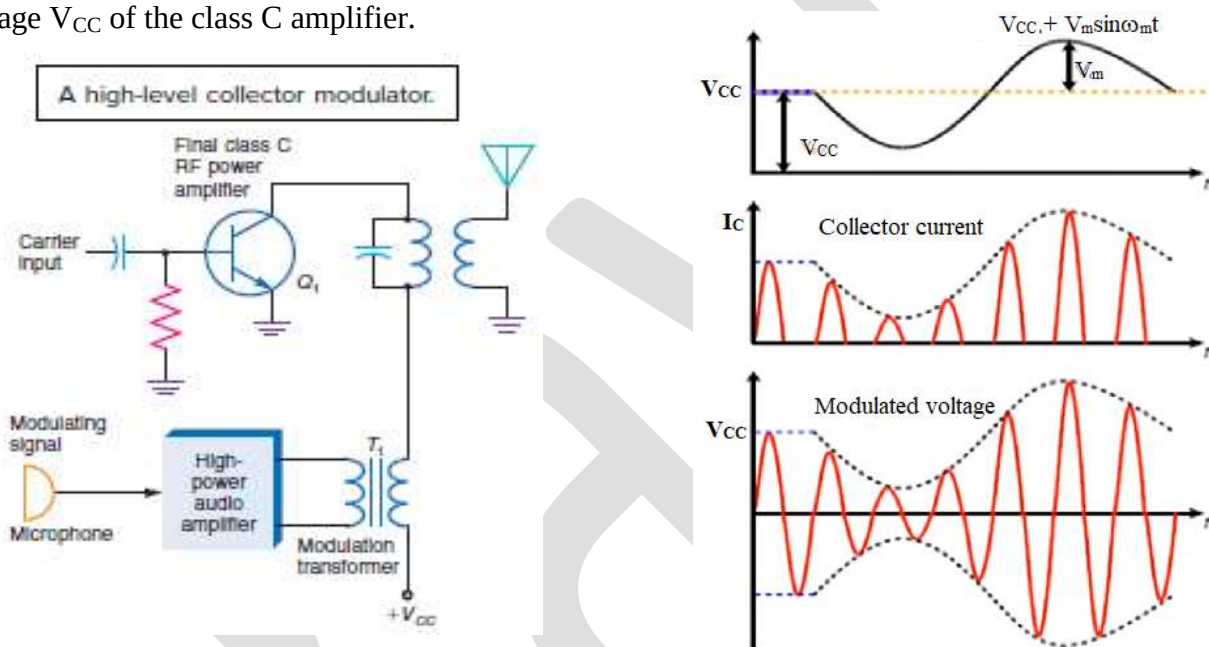
These positive going pulses are applied to the parallel tuned circuit made up of L and C with resonant frequency f_c . Each time the transistor conducts, a pulse of current flows through the tuned circuit (capacitor charges to the peak value of the current pulse). The coil and capacitor repeatedly exchange energy when the diode is off causing an oscillation at the resonant frequency f_c . The resulting output wave form across the tuned circuit is a classic amplified version of AM signal.

High-Level AM

In high-level AM, the modulator varies the voltage and power in the final RF amplifier stage of the transmitter. The result is high efficiency in the RF amplifier and overall high-quality performance.

12) With a neat circuit diagram explain the working of high level Collector modulator circuit in detail

The modulator is a linear power amplifier that takes the low-level modulating signal and amplifies it to a high power level. The modulating output signal is coupled through modulation transformer T_1 to the class C amplifier. The secondary winding of the modulation transformer is connected in series with the collector supply voltage V_{CC} of the class C amplifier.



In the absence of modulating signal, there is zero modulation voltage across the secondary of T_1 , the collector supply voltage is applied directly to the class C amplifier, and the output carrier is a steady sine wave.

When the modulating signal is applied, the ac voltage of the modulating signal across the secondary of the modulation transformer T_1 is added to and subtracted from the dc collector supply voltage. This varying supply voltage is then applied to the class C amplifier, causing the amplitude of the current pulses through transistor Q_1 to vary. As a result, the amplitude of the carrier sine wave varies in accordance with the modulated signal.

- When the modulation signal goes positive, it adds to the collector supply voltage, thereby increasing its value and causing higher current pulses and a higher-amplitude carrier.
- When the modulating signal goes negative, it subtracts from the collector supply voltage, decreasing it. For that reason, the class C amplifier current pulses are smaller, resulting in a lower-amplitude carrier output.
- During the negative cycles of carrier, the transistor turns off allowing the tuned circuit to exchange energy between coil and capacitor causing oscillations resulting in a complete AM wave at the output which will be radiated through the antenna.

Demodulation and Amplitude Demodulators

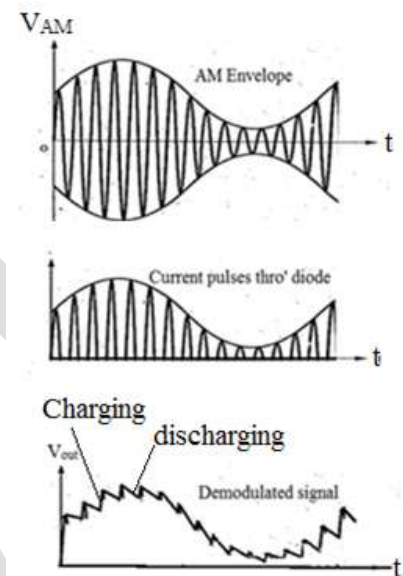
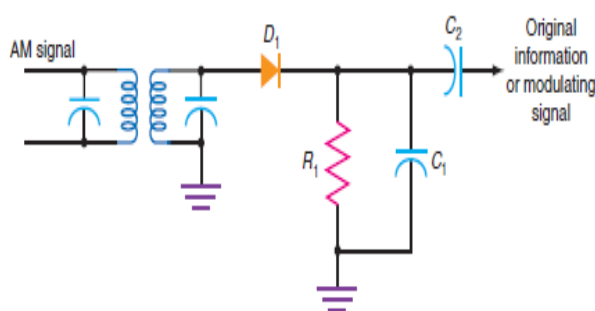
Demodulation or detection is the process of recovering the original message signal from the modulated wave at the receiver. It is the inverse of modulation process. Using a non linear device or by using an envelope detector the original signal can be recovered from the modulated envelope.

Demodulators or Detectors, are circuits that accept modulated signals and recover the original modulating signal.

Diode Demodulator or Envelope Detector

13) With a neat circuit diagrams and waveforms, explain the operation of diode demodulator.

A diode demodulator or an Envelope detector is a simple and highly effective device used to demodulate the AM wave. It consists of diode and a resistor-capacitor (RC) filter



Operation:

During positive half cycle of the input signal, the diode is forward biased and the capacitor C_1 charges up to the peak value of the input signal. When the input voltage falls below this value, the diode becomes reverse biased and now the capacitor C discharges slowly through the load R_1 . As a result only positive half cycle of the AM wave appears at the output. The discharging process continues until the next positive half cycle.

When the input voltage exceeds the voltage across the capacitor in the next positive cycle, the diode starts conducting, the capacitor charges to the next peak and the process repeats.

For proper demodulation, the time constant of the RC circuit should satisfy $\frac{1}{f_c} \ll R_1 C_1 \ll \frac{1}{2f_m}$

where $2f_m$ is the message bandwidth. The output is nearly same as the envelope with small ripple and this ripple can be easily removed using a LPF.

Generation of DSB-SC signals - Balance Modulator (Lattice or Ring Modulator)

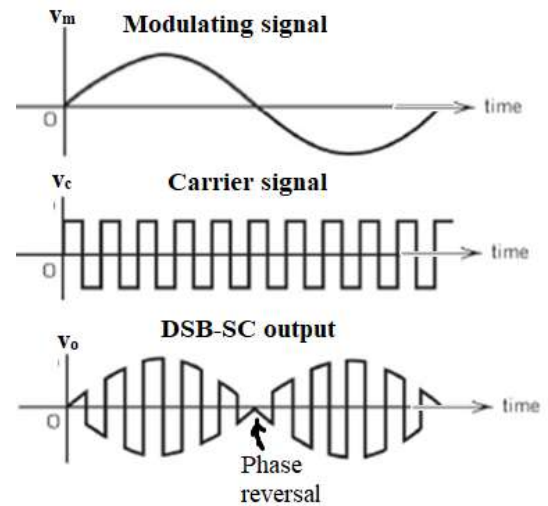
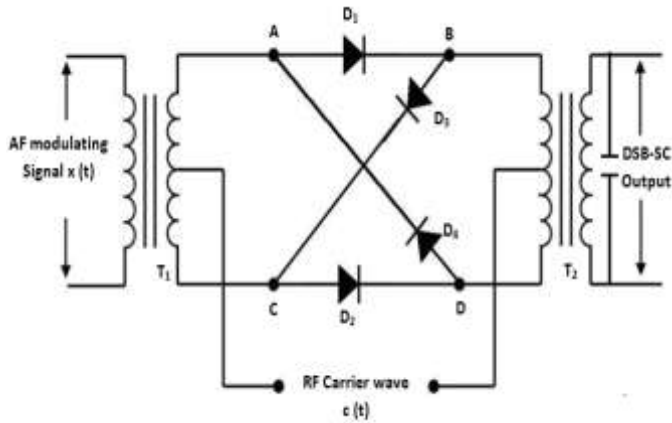
A balanced modulator is a circuit that generates a DSB signal, suppressing the carrier and leaving only the sum and difference frequencies at the output. The output of a balanced modulator can be further processed by filters or phase-shifting circuitry to eliminate one of the sidebands, resulting in an SSB signal.

14) With a neat circuit diagram and waveforms, explain the operation of ring modulator

Construction - A Balanced modulator is basically a diode ring consisting of an input transformer T_1 , an output transformer T_2 , and four diodes connected in a bridge circuit. The carrier signal is applied to the center taps of the input and output transformers, and the modulating signal is applied to the input transformer T_1 . The output appears across the secondary of the output transformer T_2 .

Carrier Suppression

Let us assume that the modulating signal is absent and only the carrier signal is applied. Hence $x(t) = 0$. During the positive half cycle of the carrier, the diodes D_1 and D_2 are forward biased and the diodes D_3 and D_4 are reverse biased.

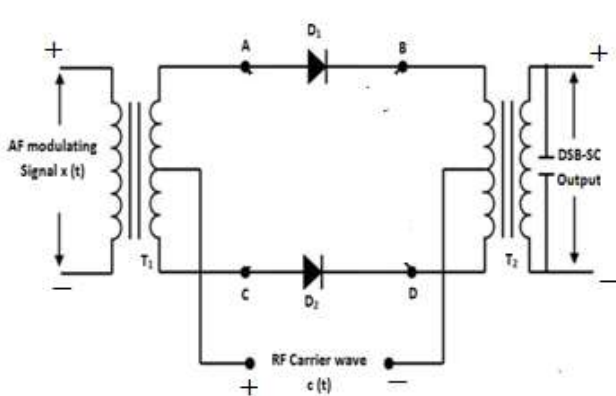


During the negative half cycle of the carrier, the diodes D_3 and D_4 are forward biased and the diodes D_1 and D_2 are reverse biased.

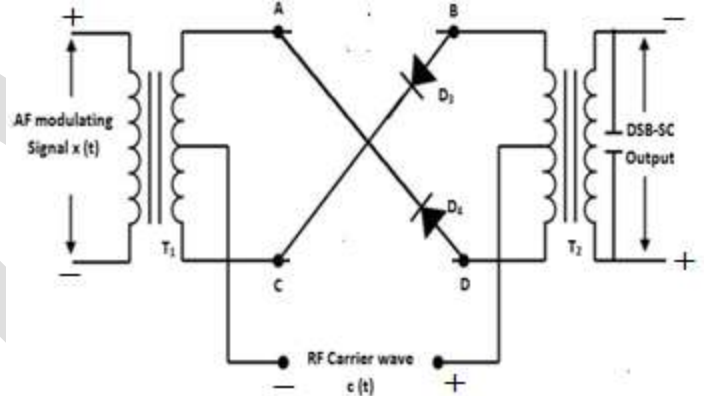
The direction of currents flowing through the primary windings of output transformer T_2 are equal and opposite to each other. Therefore, the magnetic fields produced by these currents are equal and opposite and cancel each other. Hence, the induced voltage in secondary winding is zero. Thus, the carrier is suppressed in both positive and negative half-cycles of the carrier.

During positive half cycle of the modulating signal

Apply the low frequency modulating signal through the input audio transformer T_1



In the positive half-cycle of the carrier, D_1 and D_2 are ON and secondary of T_1 is applied as it is across the primary of T_2 . Hence, during the positive half cycle of carrier, the output of T_2 is positive.



In the negative half-cycle of the carrier, D_3 and D_4 are ON and secondary of T_1 is applied as it is across the primary of T_2 . Hence, during the negative half cycle of carrier, the output of T_2 is negative.

During Negative half cycle of the modulating signal

Apply the low frequency modulating signal through the input audio transformer T_1

In the positive half-cycle of the carrier, D_1 and D_2 are ON and secondary of T_1 is applied as it is across the primary of T_2 . Hence the output of T_2 is negative.

In the negative half-cycle of the carrier, D_3 and D_4 are ON and secondary of T_1 is applied as it is across the primary of T_2 . Hence the output of T_2 is positive.

The complete operation is tabulated below

Sl No	$x(t)$	$c(t)$	Diodes	Output
1	Positive	Positive	D_1 and D_2 Conducts	Positive
2	Positive	Negative	D_3 and D_4 Conducts	Negative
3	Negative	Positive	D_1 and D_2 Conducts	Negative
4	Negative	Negative	D_3 and D_4 Conducts	Positive

Multiplexing

15) What is Multiplexing? Mention the types and few applications.

Multiplexing is the process of simultaneously transmitting two or more individual signals over a single communication channel. Multiplexing is a signal processing operation which allows multiple independent signals to be combined into a composite signal suitable for transmission over a common channel.

Multiplexing requires that the signals are kept apart so that they can be separated at the receiving end. This can be accomplished by separating the signals either in frequency or in time.

The technique of separating the signals in frequency is referred to as Frequency Division Multiplexing (FDM).

The technique of separating the signals in time is referred to as Time Division Multiplexing (TDM).

Multiplexing is accomplished by an electronic circuit known as a multiplexer

Applications – Telephony, Telemetry, Satellite communication, Commercial FM, TV broadcast etc.

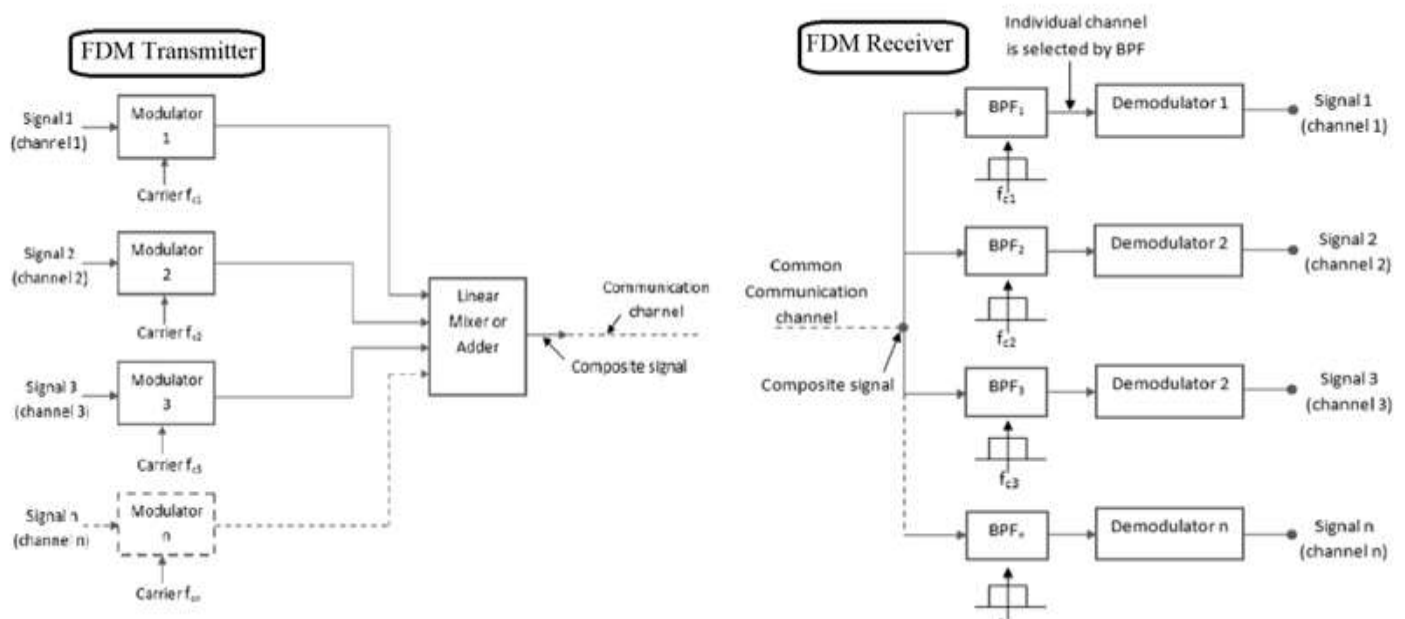
Frequency-Division Multiplexing

16) Explain the general block diagram of FDM system.

In Frequency Division Multiplexing (FDM), multiple signals share the bandwidth of a common communication channel.

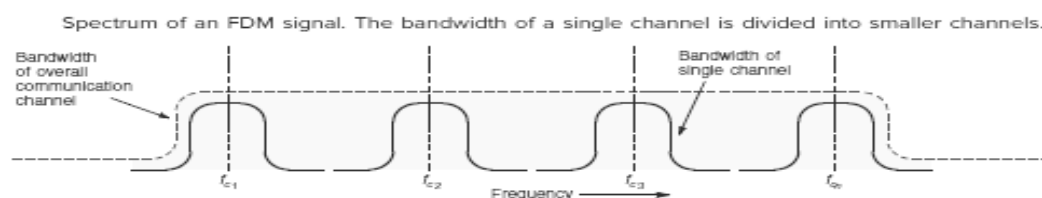
Transmitter-Multiplexers

Each signal to be transmitted feeds a modulator circuit. The carrier for each modulator (f_c) is on a different frequency. The carrier frequencies are usually equally spaced from one another over a specific frequency range.



These carriers are referred to as subcarriers. Each input signal is given a portion of the bandwidth. Any of the standard kinds of modulation can be used, including AM, SSB, FM, PM, or any of the various digital modulation methods. The FDM process divides up the bandwidth of the single channel into smaller, equally spaced channels, each capable of carrying information in sidebands. The modulator outputs containing the sideband information are added algebraically in a linear mixer.

The resulting output signal is a composite of all the modulated subcarriers. This signal can be used to modulate a radio transmitter or can itself be transmitted over the single communication channel. Alternatively, the composite signal can become one input to another multiplexed system.



Receiver- De-Multiplexers

A receiver picks up the composite signal from the communication channel and sent it to a group of bandpass filters, each centered on one of the carrier frequencies. Each filter passes only its channel and rejects all others. A channel demodulator, then recovers each original input signal.

Transmission Bandwidth

Consider an FDM system using SSB modulation to transmit 24 independent voice inputs. Assume a bandwidth of 4 kHz for each voice input. The required bandwidth for FDM is $BW = n f_m$

Where n represents the number voice signals. For 24 voice channels, $BW = 24 (4\text{kHz}) = 96 \text{ kHz}$

Advantages:

(a) Large number channels can be transmitted simultaneously. (b) FDM operation does not require strict synchronisation between transmitter and receiver and signals can be recovered easily

Disadvantages: (a) The communication channel requires a large bandwidth. (b) Problem of cross talk and wideband fading exists. (c) The system is complex as large number of filters, modulators/demodulators are required.

Numerical Problems

1) An AM transmitter uses high-level modulation of the final RF power amplifier, which has a dc supply voltage V_{CC} of 48 V with a total current I of 3.5 A. The efficiency is 70 percent. (a). What is the RF input power to the final stage? (b). How much AF power is required for 100 percent modulation? (c.) What is the carrier output power? (d.) What is the power in one sideband for 67 percent modulation? (e.) What is the maximum and minimum dc supply voltage swing with 100 percent modulation? (Model QP2 - 8M)

Soln - (a) DC input power $P_i = V_{CC} I = 48(3.5) = 168\text{W}$

(b) For 100% modulation AF power $P_m = 0.5P_i = 0.5(168) = 84 \text{ W}$

(c) Given efficiency $\eta = 70\%$ $\rightarrow P_{out} = \eta(P_i) = 0.7(168) = 117.6 \text{ W}$

(d) Given $m = 0.67$, $P_C = 168\text{W}$; Sideband Power $P_{LSB} = P_{USB} = \left(\frac{m^2}{4}\right) P_C = \left(\frac{0.67^2}{4}\right) 168 = 18.85 \text{ W}$

(e) Minimum Swing = 0; maximum Swing = $2 \times V_{CC} = 2(48) = 96 \text{ V}$

Exercise Problems (Text 2)

1) A Collector modulated transmitter has a supply voltage of 48V and an average collector current of 600 mA. What is the input power to the transmitter? How much modulating signal power is needed to produce 100 percent modulation?

Solution – Given $V_{CC} = 48\text{V}$; $I_{DC} = 600 \text{ mA}$

The DC input power $P_{in} = V_{CC} I_{DC} = 28.8\text{W}$

For 100% modulation, modulating signal power has to be 50% of input power.

Hence $P_m = 0.5(P_{in}) = 0.5(28.8) = 14.4\text{W}$

2. An SSB generator has a 9-MHz carrier and is used to pass voice frequencies in the 300- to 3300-Hz range. The lower sideband is selected. What is the approximate center frequency of the filter needed to pass the lower sideband?

Solution- Given $f_C = 9000 \text{ kHz}$ $f_{LSB1} = 9000 - 0.3 = 8999.7 \text{ kHz}$ and $f_{LSB2} = 9000 - 3.3 = 8996.7 \text{ kHz}$

The center frequency of the filter = $(8999.7 + 8996.7)/2 = 8998.2 \text{ kHz} = 8.9982 \text{ MHz}$